

Supplementary Material

S1. Why the musculoskeletal replication is out of reach

tcds_m5_myoarm.json stores 4 seeds $\times$ 6 held-out bodies. The paired difference joint – staged, in mm, by seed:

seed	held-out bodies	seed mean
0	-7.64 +8.39 -9.10 +22.13 -3.26 +7.64	+3.03
1	-10.70 -22.24 -12.46 -20.10 -6.93 -11.70	-14.02
2	-8.35 -5.91 -16.73 -1.68 +1.08 -20.38	-8.66
3	-9.38 +18.03 +61.10 +7.41 +7.91 +37.51	+20.43

Grand mean +0.193 mm. Between-seed variance of the seed means is 232.68 mm² (SD 15.25 mm); within-seed variance across bodies is 220.47 mm², contributing 220.47/6 = 36.75 mm² to a six-body seed mean. The residue, 195.93 mm², is **84% of the between-seed variance** and is training-run variance — the divergence of two independently optimised trajectories.

Minimum detectable effect, $MDE = 2.93 \cdot SD/\sqrt{n}$, under each remedy:

bodies	4 seeds	20 seeds	50 seeds
6	22.35	9.99	6.32
12	21.45	9.59	6.07
24	20.98	9.38	5.93

Adding bodies is nearly useless because the variance it reduces is only 16% of the total. Pairing more tightly is impossible: `train_joint` and `train_staged` (`redescribe_v3.py:62,88`) already call `torch.manual_seed(seed)` on the same `bodies_data`, so initialisation and data are shared; the remaining divergence is what the two curricula *are*. Reaching the 0.90 mm effect measured on `arm2` would need $n \approx 50 \cdot (5.93/0.90)^2 \approx 2170$ seeds. That figure assumes a staging effect transfers in absolute millimetres. `myoArm`'s error scale is about an order of magnitude above `arm2`'s, so if the effect instead scales with baseline error the target is nearer 9 mm and the requirement falls to tens of seeds. We cannot distinguish the two, which is the sense in which the round is uninformative.

S2. The two attempts on the control objective

First attempt. Treatment: a training-time curvature penalty on the code ($\lambda = 3.0$, selected on seeds disjoint from the test by a rule fixed in advance). Control: a code-jitter arm, smooth in θ without imposing linearity. Noise floor: a second identically-trained model differing only in its init seed. The penalty bit as intended — curvature cut to 0.26–0.57 of baseline on all 12 seeds, at a one-step fidelity cost under 1.5% — but the quantity under test was a difference of differences, 2.3 mm against 54 mm of MPC sampling noise and a 4.2 mm noise floor. Three of five checks failed. The misallocation was ours: compute went to seeds rather than to restarts and shooting repetitions.

Second attempt. The pre-registration made power an *entry condition* rather than a post-hoc check: the test was not permitted to run unless the projected minimum detectable effect fell under 8 mm. Averaging sixteen shooting seeds into the metric cut noise from 54 mm to 21 mm on the calibration block and projected an MDE of 6.2 mm, so the round opened. The noise-floor check passed this time (2.24 mm against a 4.45 mm treatment effect). The attempt nonetheless failed for three reasons. On the fresh, disjoint test block, adapting the code did not beat leaving it unadapted on the pre-registered mean-over-restarts metric (234.4 mm against 232.9 mm); one seed's curvature ratio

fell to 0.188, below the pre-set 0.2 floor; and the **achieved** MDE was 9.05 mm with 25.0 mm of noise — so applying the criterion to the realised run rather than to the projection would not have passed it.

On a best-of-restarts statistic the Section IV-C4 round and this second attempt give 104.1 mm against 143.3 mm unadapted and 212.9 mm against 232.9 mm. We read no direction from either: the adapted arm is the minimum of eight restarts and the unadapted arm a single evaluation, so the comparison is confounded by the restart count, which is why Section IV-C4 withdraws check 1 rather than scoring it. What the two rounds do differ in is the seed block's absolute difficulty, by a factor of 2.0. The unadapted arm is not restarted, so 232.9 mm is its value under either statistic. A 4.5 mm treatment effect cannot be read against a baseline that moves by more than a hundred. The pre-registration required us to stop after a second failure.

S3. The two nonlinear re-encodings and the gate

`scrambled` is a four-layer additive coupling of the RealNVP form. Write the code as two halves (a, b) of two dimensions each; layer k then replaces b by $b + \tanh((a/\sigma_k)W_1 + b_1)W_2\sigma'_k$ and then swaps the halves. The layer count is even so the swaps cancel and the map returns the coordinates in their original order, and which half conditions layer k is tracked by the layer index and not inferred from tensor width. With an even code dimension both halves are the same width, so inference would pick the wrong normaliser. W_1 is 2×16 , b_1 is 16, W_2 is 16×2 ; entries of W_1 and b_1 are drawn from $N(0, 2^2)$ and entries of W_2 from $N(0, (2/\sqrt{16})^2)$. The four layers do not share weights. Each draws its own triple, in layer order, from one generator seeded by the run's seed. σ_k is the per-dimension standard deviation of that seed's ten training codes over the conditioning half, and σ'_k the same over the half being displaced. Both are computed once, from the original training codes, and reused at every layer. The subscript selects which half, not which layer's inputs. (For the gated map the distinction is empty, since the training codes are fixed points of the map. For `scrambled` it is not, because the codes move through layers 1–3, where σ sets the scale of the whole displacement.) Without that scaling the map is nonlinear in name only: θ sits near the origin where $\tanh$ is nearly linear, and a nominally nonlinear map is affine in disguise, which is what our first smoke test found.

`nonlinear_readable`, the *gated* arm, reuses the same weights and multiplies layer k's displacement by

$$g(a) = \prod_{j=1..10} [1 - \exp(-\|a - A_j\|^2 / 2\sigma_g^2)],$$

the product taken over the ten training codes A_j restricted to that layer's conditioning half. The gate is a function of the conditioning half alone, so each layer remains an exactly invertible additive coupling and the wrapper's inverse is analytic. The round-trip is verified to 10^{-6} per seed. The gate is exactly zero at every training code, so each layer, and hence the composition, is the identity there. Leave-one-body-out R^2 is a function of the training codes alone, so decodability is preserved to machine precision *by construction* while the map stays nonlinear away from those codes, which is where adaptation runs; away from the anchors the gated and scrambled maps are the same map. The width is $\sigma_g = 0.12 \times$ the median nearest-neighbour Euclidean distance among the ten training codes, measured in the full four-dimensional θ and then applied to two-dimensional half-code distances. It was fixed by calibration over $\{0.12, 0.18, 0.25\}$, choosing the value that leaves the gate open, above one half, on the largest fraction of 64 probe codes, drawn from a standard normal with a fixed generator and rescaled to the training codes' per-dimension mean and standard deviation (means over four calibration seeds: 0.88, 0.75, 0.56). That rule is monotone by construction and carries no information. The gate is strictly decreasing in σ_g at fixed probe codes, so "the largest open fraction" can only ever return the smallest candidate. The three measured fractions are monotone because they must be. The calibration therefore selected a grid endpoint, and since a smaller width makes the map more nonlinear over the tested range, the rule is biased towards a larger reported cost. It did not in fact select the maximum: Section IV-C1 measures a marginally higher cost at 0.18. Section IV-C1 therefore reports the gated map at all three calibrated widths. The calibration was also not pre-registered. Its runs finished 39 to 121 seconds before the document that records the selection rule. The rule therefore predates the Section IV-C1 experiment it governs, but not the calibration data that motivated it, and the width should be read as a tuned hyper-parameter.

organised is the identity. $k1$ and $k300$ are linear, $\theta \mapsto \theta A^T$ with $A = U \text{diag}(s) V^T$ for independent random orthogonal U and V and $s_i = \kappa^{e_i}$, the exponents e evenly spaced on $[-\frac{1}{2}, +\frac{1}{2}]$. The singular values are therefore log-spaced with geometric mean 1 and ratio $\kappa \in \{1, 300\}$; fixing the geometric mean keeps κ from also changing the map's overall scale, which would confound conditioning with magnitude.

S4. Gate-width sensitivity

The width was chosen by a rule that is monotone by construction and could only ever have returned the smallest candidate (Section III), so we re-ran the whole experiment at the two widths the rule could not have chosen. The gated map costs +0.875 mm at 0.12 ([+0.474, +1.275]), +0.878 mm at 0.18 ([+0.477, +1.280]) and +0.733 mm at 0.25 ([+0.356, +1.110]), all at $n = 20$ with $p \leq 6.5 \times 10^{-4}$. The ungated scrambled arm, which has no width, is +0.653 mm in all three runs, which is the check that nothing else moved. The construction's properties hold throughout. Leave-one-body-out R^2 is preserved exactly at every width and seed (gap 0.000), the gate is exactly zero at every anchor, and the Jacobian condition number is 90.9, 88.5 and 84.5, so the width changes how much of the space the map bends and not how badly it is conditioned. The open fraction falls monotonically across the twenty decisive seeds (0.90, 0.78, 0.63), as the rule's monotonicity requires. Because the three runs share seeds, bodies, budgets and starts, the ungated map is identical seed by seed in all three. The statement about stability therefore belongs at the paired level, and at that level no pair of widths is distinguishable: 0.12 – 0.18 is –0.004 mm ([–0.269, +0.262], $p = 0.98$), 0.12 – 0.25 is +0.142 mm ([–0.106, +0.389], $p = 0.25$) and 0.18 – 0.25 is +0.145 mm ([–0.142, +0.432], $p = 0.30$). Those intervals exclude width differences larger than about 0.4 mm against an effect of 0.875 mm. The +0.875 mm quoted in the abstract is the 0.12 result, reported because it is the one the calibration selected, not because it is the largest: 0.18 is marginally larger. Three widths on a bounded interval are not a curve, and we draw none. The cost must fall to the ungated +0.653 mm as the gate opens everywhere and to zero as it closes everywhere, so these points sit on a hump whose shape we have not measured. Widths below 0.12 are untested.

S5. Adaptation and its start

Section III states the adaptation objective and the reported metric. This section gives the update rules and the start. We use two rules. L-BFGS: `max_iter` 80, `history` 20, strong-Wolfe line search, `tolerance_grad` 10^{-12} , `tolerance_change` 10^{-14} , one step call, falling back to the start if the result is not finite. Adam: 80 steps at each of three learning rates {0.015, 0.05, 0.15}. Both run at each of three adaptation budgets {25, 100, 400} transitions, on 6 held-out bodies per seed over 20 seeds. Adaptation begins from a start θ_0 drawn once per seed as $0.1 \times N(0, I)$. Each arm begins from its own image $M\theta_0$ of that same point, which is what carries the identity to the estimate (Section III).

S6. Statistics and pre-registration coverage

Minimum detectable effects. Section III gives the definition. The spread is the round's own. The constant is not, so an MDE quoted for a four-seed round is smaller, by the ratio of critical values (4.02/2.93), than the one that round's own critical value would give, and we quote the smaller figure throughout. The ten per-setting minimum detectable effects of Section IV-C4 are *projected* to $n = 20$ and marked as projections wherever they appear; Section IV-C4's own start-point spreads are measured at that n . Section VI's myoArm figure is the achieved value at the 4 seeds $\times$ 6 held-out bodies that round ran. S1's table gives that same cell plus counterfactual MDEs at seed and body counts we did not run, and Section VI quotes two of them as projections. S2's control-metric entry criterion is a third case and we flag it rather than fold it in. Its entry condition was an MDE of 6.2 mm *projected from a calibration block*, against 9.05 mm achieved on the test block, and we keep projected and achieved values distinct wherever either is quoted.

Unit of analysis. The unit of analysis is the seed, because that is the level at which every manipulation is applied. Where an earlier analysis bootstrapped over held-out bodies or cells we report the seed-level interval instead. That is not uniformly a widening: mechanism 5's seed-level interval is 1.9 $\times$ the width of its 30-cell bootstrap, while mechanism 4's is slightly narrower.

Manipulation checks. Section IV-A and Section IV-C each contain an instance of the rule Section III states. Each pre-registered round's manipulation checks and decision rule were fixed in advance, in most cases together with the conclusion paragraph for every possible outcome, so that the run could only select one. The documents are in the repository.

Pre-registration coverage. These are the results Section III names as uncovered. The five-seed staged-versus-joint comparison in Section IV-A, the staged-versus-joint figures closing Section IV-B, and both of Section IV-D's control metrics predate the project's first design document by three weeks; Section IV-B's twelve-seed decodability result postdates it but carries no design of its own. These are exploratory, and each is marked as such where it is reported. Only the staging comparison was subsequently re-run under a pre-registration written before its script first executed, at twenty fresh seeds on three bodies, two of which passed the substituted manipulation check (Section IV-A), and it is that replication which the abstract reports; Section IV-A gives the exploratory run first and the replication after it. The difficulty round's four uncovered settings are the fourth case (Section III).

S7. The whitening moderator analysis

This is the moderator Section IV-C2 reports, in full. Regressing each seed's WHITE – ORIG contrast on the log condition number of that seed's training-code covariance over the 20 seeds gives a slope of -0.063 mm per log unit (95% CI $[-0.170, +0.044]$, $p = 0.23$, Spearman $\rho = -0.23$), and the worst-conditioned half of seeds averages -0.025 mm against $+0.074$ mm for the best half. The sign is the one the dilution hypothesis predicts and the effect is not resolved. Taken at face value across the whole observed range it would come to about -0.26 mm, still a fifth of the $+1.234$ mm the instrument does resolve. Under L-BFGS the same regression is flat ($+0.0001$ mm per log unit, $p = 0.51$), as the identity requires (`src/whitening_moderator.py`).

S8. Budget-matched staging contrasts

DESIGN_MATCH.md fixes five training conditions in advance: the published joint arm, a joint arm cut to 19 epochs (1121 steps against the staged arm's 1150), and three staged arms whose weighted budgets are 40%, 50% and 69% of the joint arm's. The direction survives on both bodies at every budget, but the magnitude attenuates on arm3:

body	staged arm's weighted budget	joint – staged	seed-level CI	p
arm2	40%	-1.150 mm	$[-1.713, -0.586]$	4.1×10^{-4}
arm2	50%	-0.694 mm	$[-1.201, -0.187]$	0.010
arm2	69%	-1.082 mm	$[-1.607, -0.557]$	3.7×10^{-4}
arm3	40%	-1.238 mm	$[-1.706, -0.771]$	2.4×10^{-5}
arm3	50%	-0.928 mm	$[-1.381, -0.475]$	4.0×10^{-4}
arm3	69%	-0.811 mm	$[-1.151, -0.470]$	8.2×10^{-5}

Seventeen of twenty seeds favour joint training on arm2 at the 40% budget, and eighteen of twenty on arm3. On arm3 the effect falls monotonically as the budget closes, by 34% across the tested range. On arm2 it dips and rebounds. A fully equalised weighted budget is the point at which the curriculum stops differing from joint training, so no trend can be read as predicting an effect there. Taken only as a check on the tested range, arm3's linear trend does not reach zero inside it, and extended to a fully equalised budget it would still leave -0.35 mm. The direction holds at all three budgets on both bodies. We cannot say the effect is independent of the residual disparity. The 78% raw disparity is not a design choice: it falls out of the $\max(6, \text{epochs}/3)$ schedule arithmetic. All five training conditions share seeds, training bodies, data, held-out bodies and adaptation procedure, and the design required the staged variants to be reported together, as they are here.

A 4-seed run on arm3 gives -0.25 mm with a 24-cell bootstrap interval of $[-1.61, +1.29]$, wide enough to contain the 20-seed effect. That run is underpowered rather than a negative replication.